DEVELOPMENT OF TONGUE & SALIVARY GLANDS

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TASK: COMPLETE THE FOLLOWING QUESTIONS

Q. Give an account on development of tongue in tabular form.

Region of the Tongue	Development	Key Structures Involved	Embryological Origin
Anterior Two-Thirds (Oral Tongue)	Develops from three swellings	- Distal tongue buds (lateral lingual swellings)	First pharyngeal (branchial) arch
		- Median tongue bud (tuberculum impar)	
	- Distal tongue buds rapidly expand to overgrow median bud	- Foramen caecum (caudal border)	
	- Median tongue bud contributes little to adult tongue		
Posterior Third (Pharyngeal Tongue)	Develops from a single midline swelling	- Hypopharyngeal eminence	Mainly from third pharyngeal (branchial) arch
	- Rapidly overgrows second arch (copula)	- Merges with first arch swellings	Small contribution from fourth arch
Muscle Development	Primarily from paraxial mesoderm	- Occipital somites	

Q. How are the different regions of the adult tongue innervated?

Region of the Tongue	Nerve Supply	Function
Anterior Two-Thirds	Lingual Nerve (from the first pharyngeal arch)	General sensation
	Chorda Tympani Nerve (from the facial nerve)	Taste (considered a 'pretrematic' nerve)
Posterior Third	Glossopharyngeal Nerve (from the third pharyngeal arch)	General sensation and taste
	Superior Laryngeal Nerve (from the fourth pharyngeal arch)	Taste and general sensation
Muscles of the Tongue	Hypoglossal Nerve	Motor control

TASK: IDENTIFY, LABEL & PASTE THE FOLLOWING FIGURE

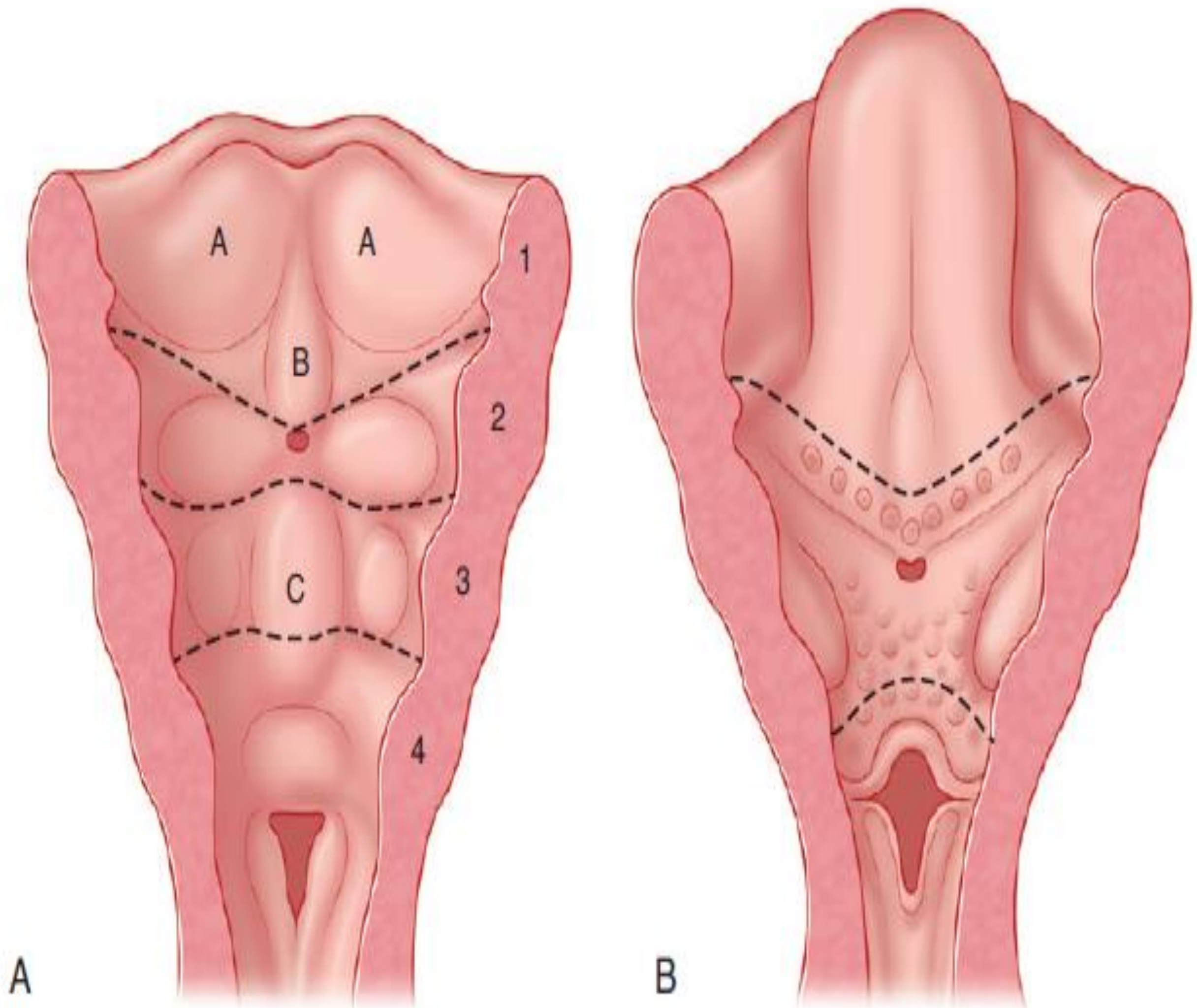


Fig. 20.1 The schema of tongue primordia. (A) The ventral wall of the pharynx in the 4th week of intra-uterine life. (B) The developing tongue at the 5th month. Numbers 1–4 indicate the positions of the pharyngeal (branchial) arches. Four swellings are seen: two distal tongue buds (lateral lingual swellings, A), the median tongue bud (tuberculum impar, B) and the hypopharyngeal eminence (C).

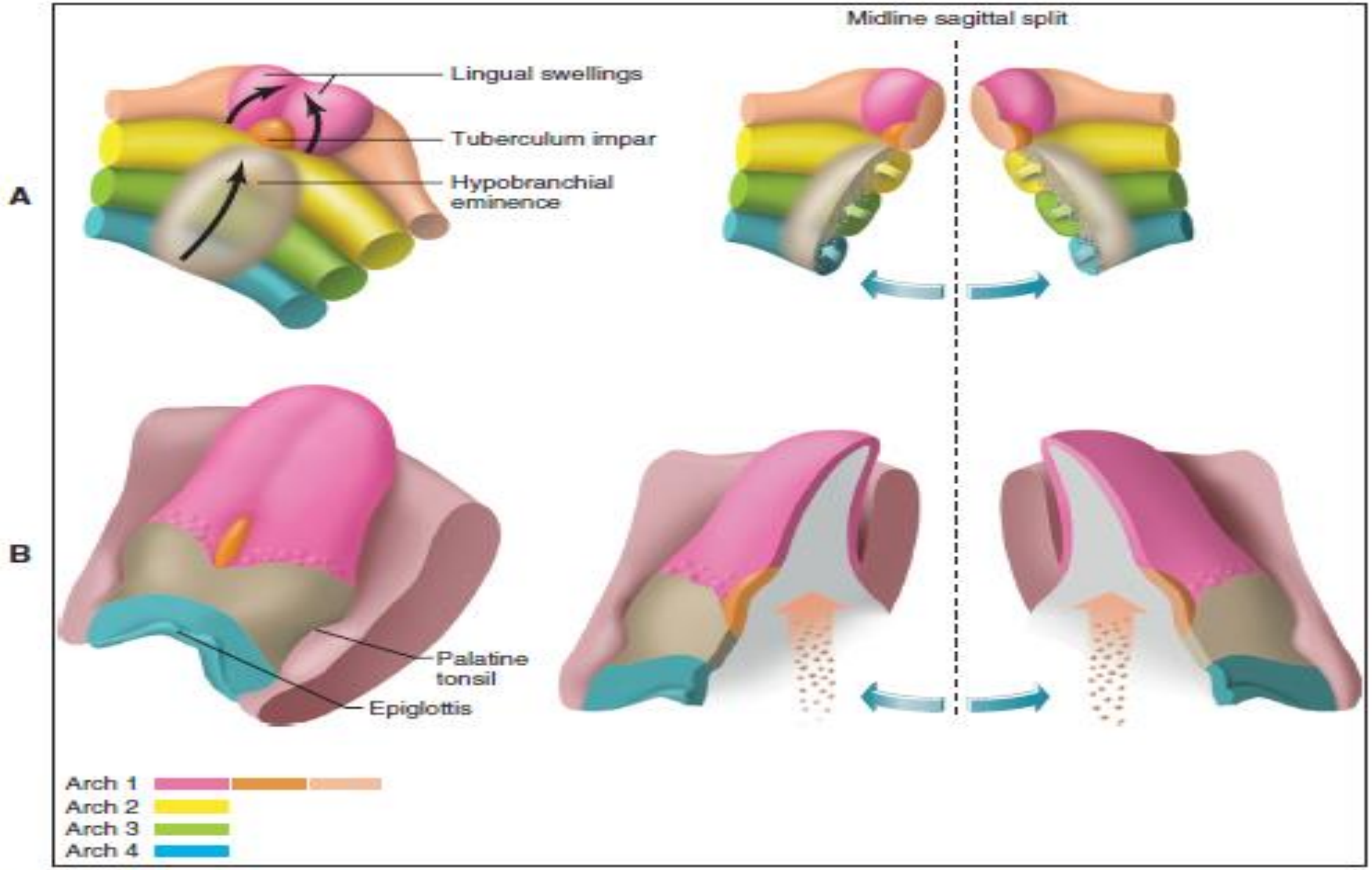
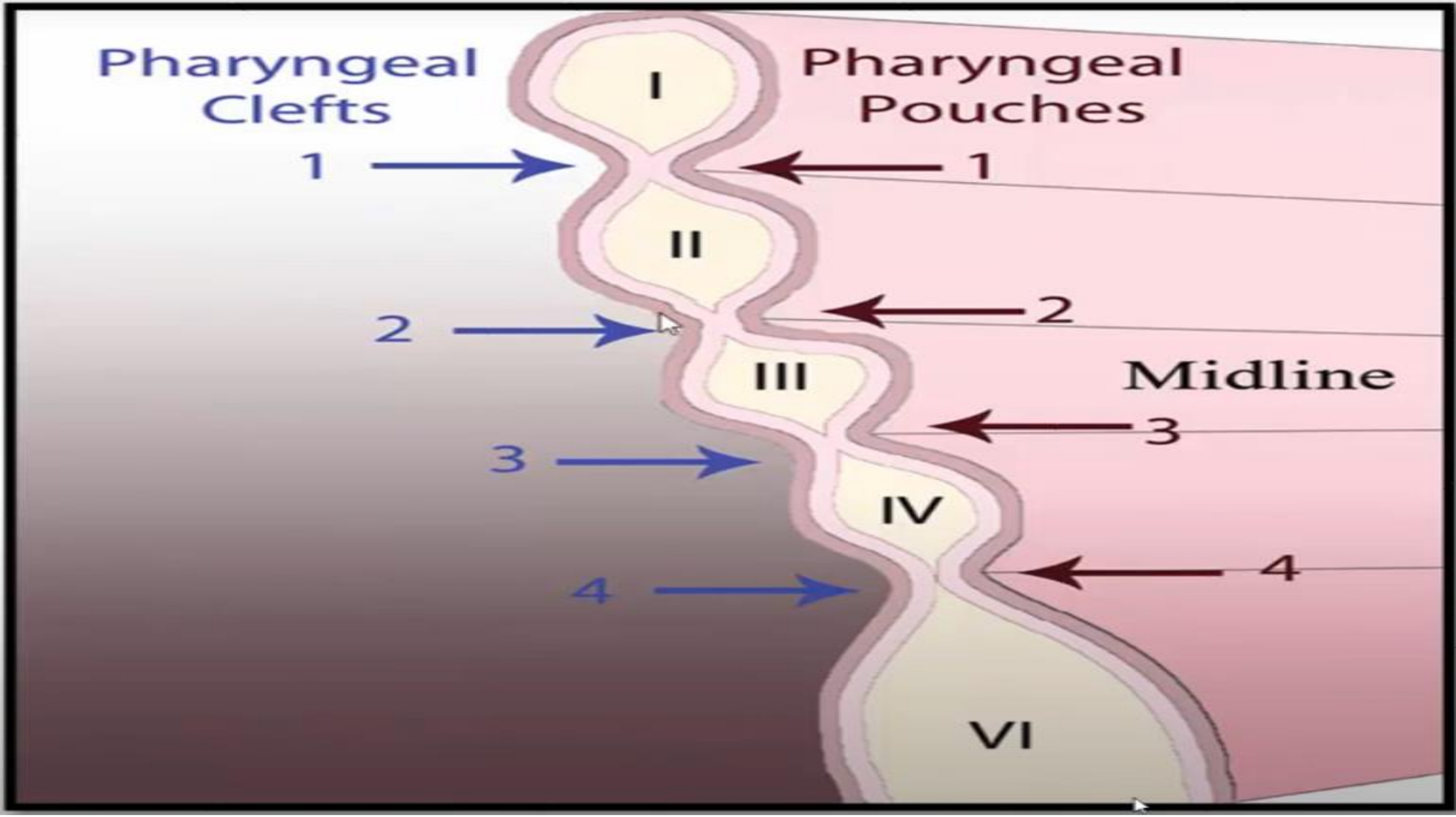
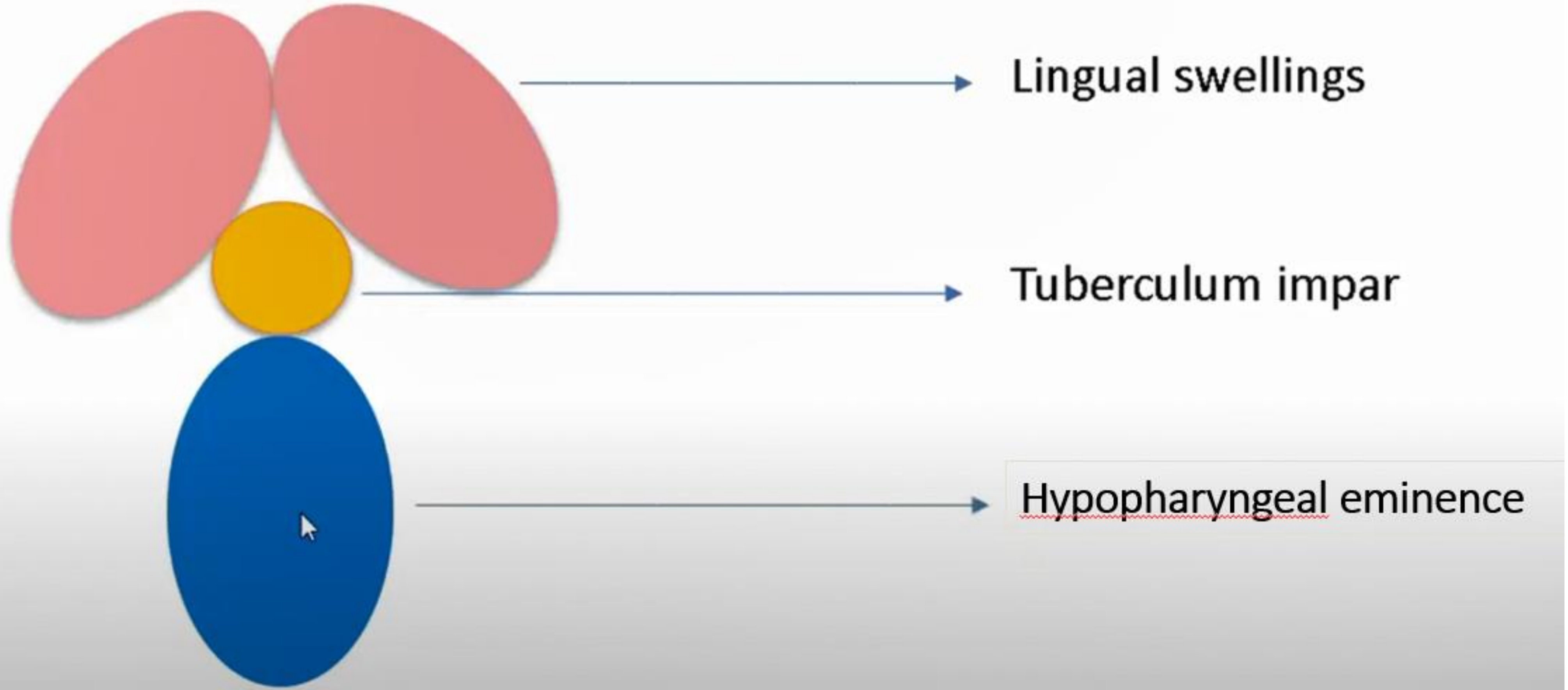


FIGURE 3-22 Development of the tongue. **A**, The lingual swellings, together with the tuberculum impar, which arise from the first arch, will form the anterior two thirds of the tongue. The hypobranchial eminence overgrows the second arch. **B**, Final disposition of the tongue and the relative contributions of the first to fourth arch. The *arrow* depicts the route of incoming occipital myotomes that form the tongue muscle.

EXTRA REFERENCE MATERIAL


Pharyngeal Clefts
Pharyngeal Pouches
Midline
1
2
3
4
I
II
III
IV
VI
1
2
3
4

Development of the Tongue

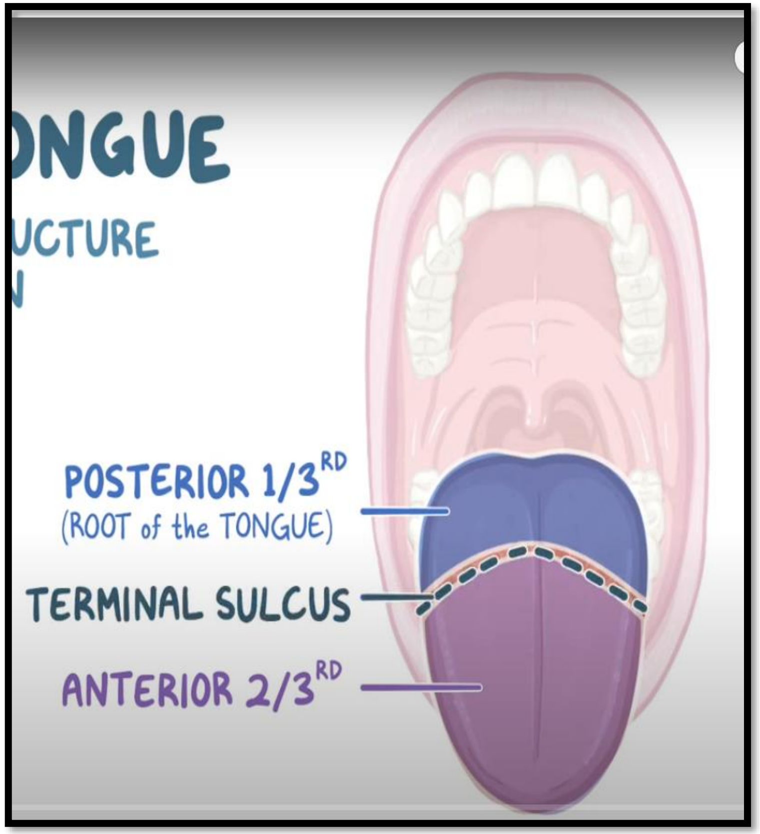


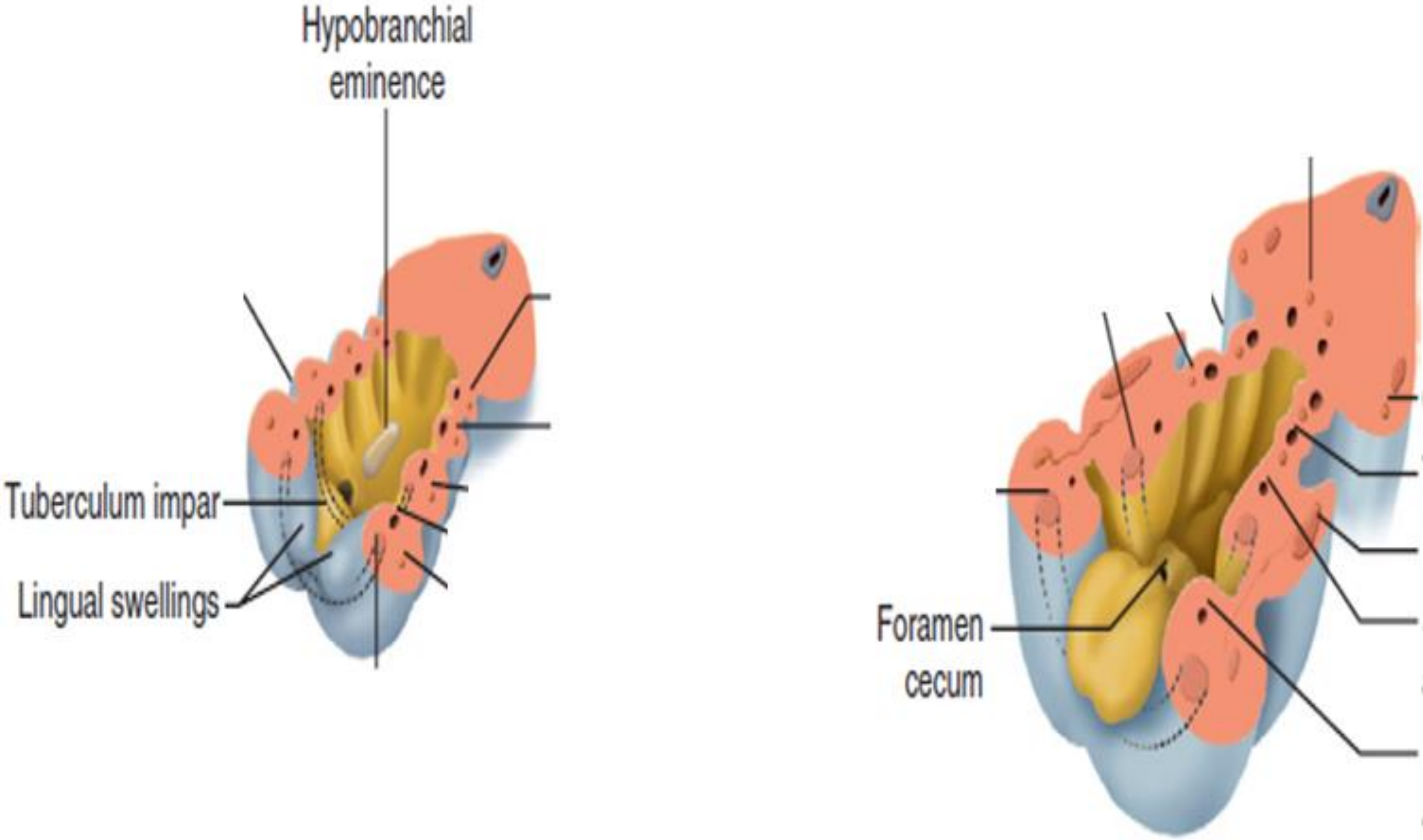
TONGUE DEVELOPMENT

Time period: Begins at about 4 weeks.

Meeting of pharyngeal arches beneath primitive mouth

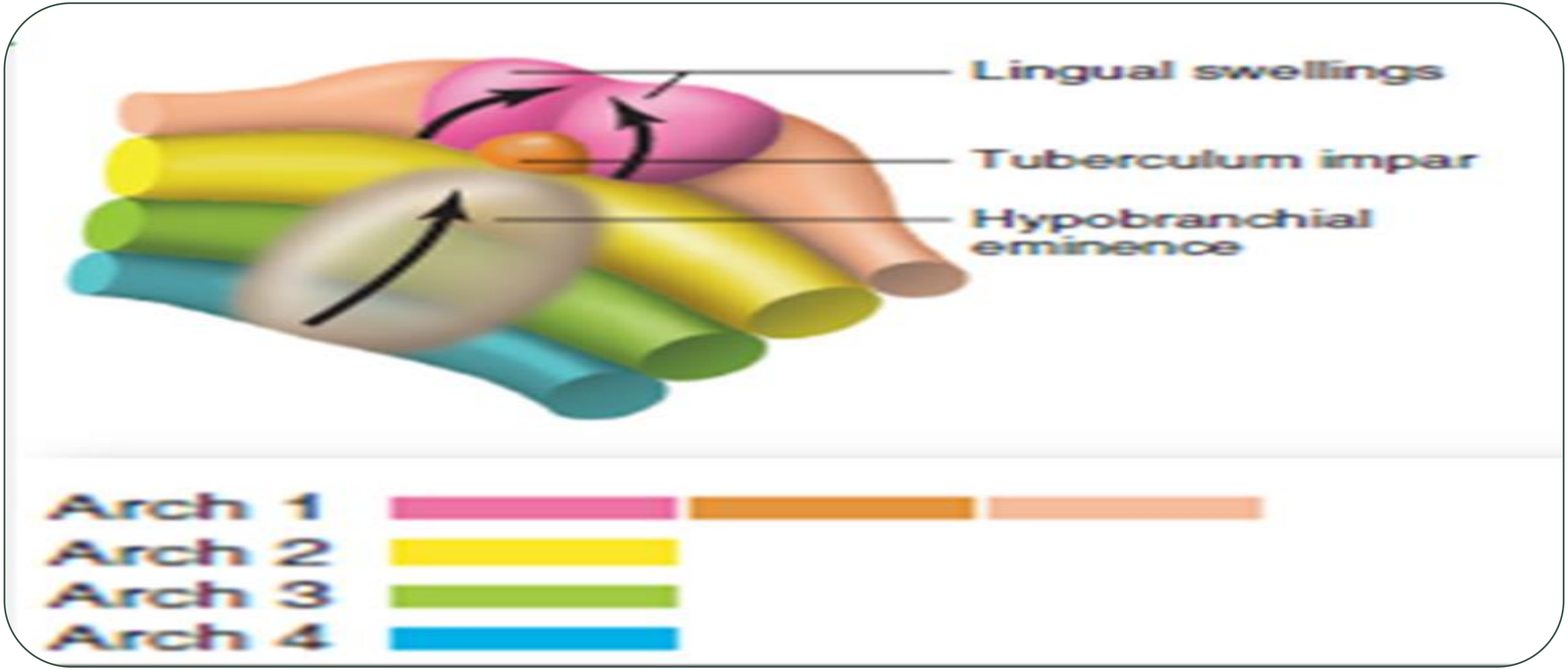
Local mesenchymal proliferation → swelling in floor of mouth



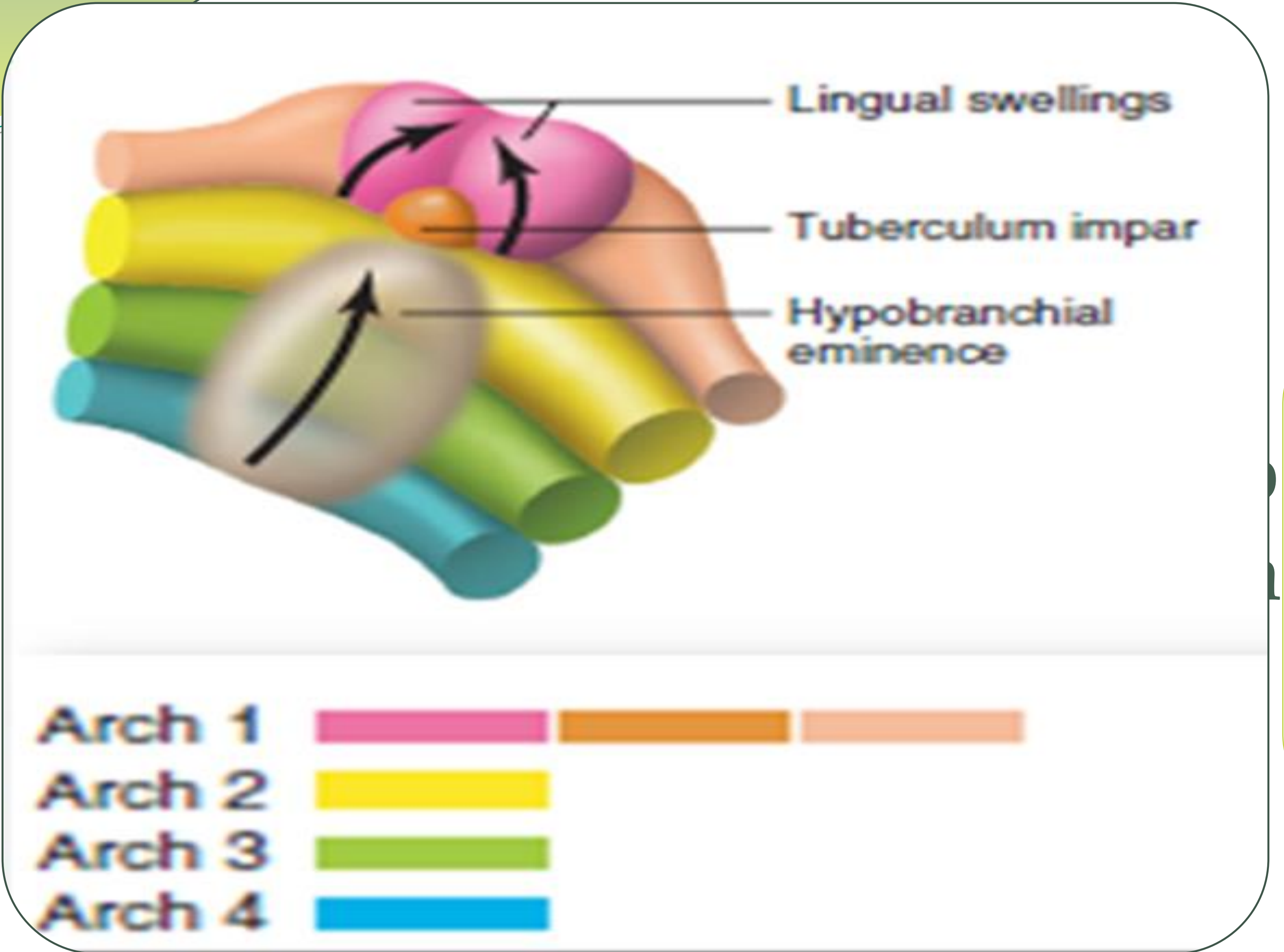


Hypobranchial eminence
Tuberculum impar
Lingual swellings
Foramen cecum

Anterior 2/3rds of Tongue (Oral Tongue):

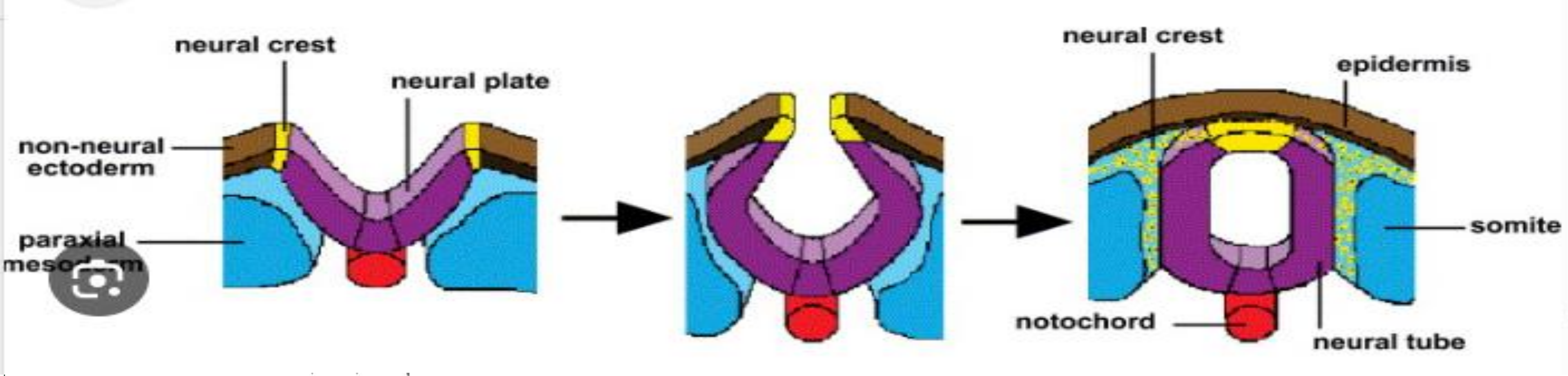


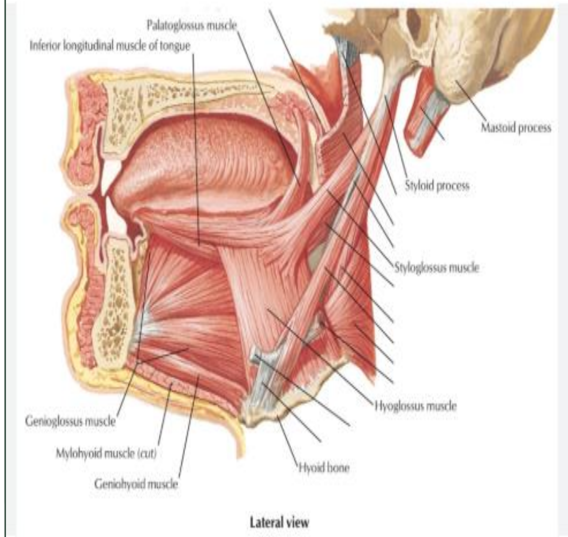
Origin: Mesenchymal proliferation beneath endodermal lining of 1st pharyngeal arch.



Posterior 1/3rd of tongue:

Hypopharyngeal eminence origin: mainly 3rd pharyngeal (branchial) arch (with a small contribution from the 4th arch).





Muscles of the Tongue:

Origin: paraxial mesoderm of the occipital somites that migrates into the developing tongue.

Instructive role of Neural Crest cells:
Smad4-mediated transforming growth factor beta (TGFβ) signaling cascade involving Fibroblast Growth Factor 6 (FGF6)

GENES AND GROWTH FACTORS INVOLVED IN DEVELOPMENT & SHAPE OF VARIOUS TONGUE PAPILLA:

Filiform Papillae

Bone Morphogenetic Proteins (BMPs)

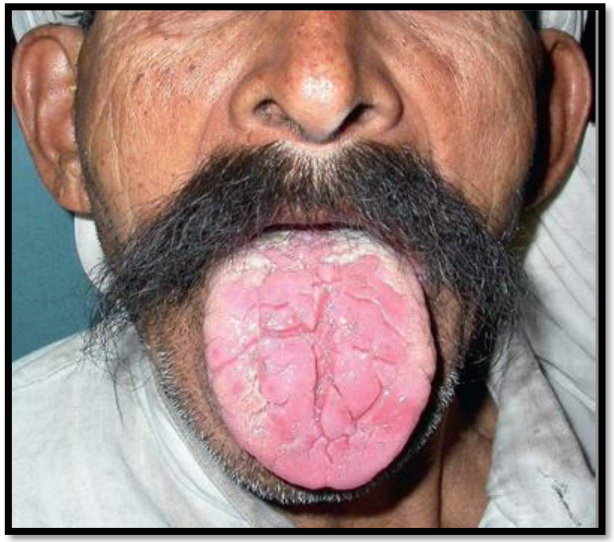
Fungiform Papillae

- sonic hedgehog (Shh)
- Wnt6
- Lef1
- Sox2
- epidermal growth factor (EGF)

Development of the taste buds,

Sox2

Lingual Frenum



Macroglossia

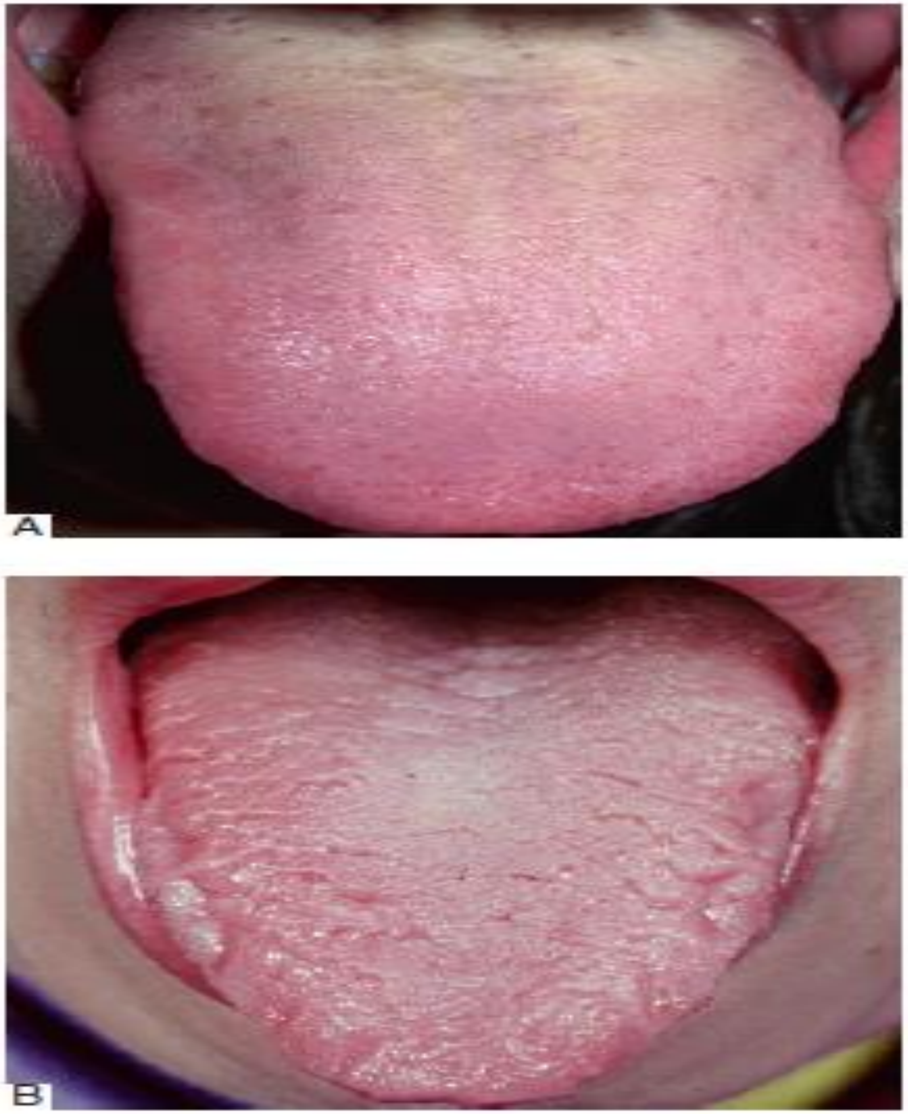


Fig. 20.5 (A) Normal tongue. The red spots represent fungiform papillae. (B) A fissured tongue where the dorsal surface is grooved. Courtesy Prof. P R Morgan.

Median Rhomboid Glossitis